CallMindVoice: an AI Voice Agent and its Financial Management

Full explanation of the project: what it is, how it works, what is connected to what, and how its money is planned and controlled

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Program	ITPU Master's Degree [specialization, years]
Internship	EPAM Systems, Inc., 7 September – 23 October 2026
Project	CallMindVoice (working name "Ovoz"), a team project of 7 people
My role	Project manager, focus on **finance**
Technical mentor	[name]
Scientific advisor	[name]
Thesis topic	Comparative Research on the Effectiveness of AI-Driven Intelligent Agent Systems and Traditional Approaches in Financial Management of IT Projects
Document date	16 October 2026 (version 1)

> **How to read this document.** It uses simple English. Hard words are explained in the glossary at the end (section 14). If you only have one minute, read section 1.

Contents

> If the contents list is empty, right-click it in Word and choose **Update field**.

1. The project in one minute

- **What:** CallMindVoice, an AI voice agent that answers customer phone calls in Uzbek, Russian and English, using only the company's own documents.
- **My role:** project manager (PM), responsible for **finance**: how much the project costs, where the money goes, and whether the product pays back.
- **The problem:** in an AI product every call costs money (phone minutes, AI, speech). Without control, the costs grow with every call, the team spends more hours than planned, and the customer cannot see the value.
- **The solution:** finance is built into the system. Every call is measured in minutes, every minute has a known cost, cheap engines are used first, and the console shows cost, savings and invoices in UZS.
- **Budget so far (estimate):** planned about 74.1 million UZS, actual about 76.0 million UZS, so about 2.5% over budget. AI and telephony were about 1.9 million UZS below plan.
- **Result so far:** the MVP was ready on 16 October 2026. It works end to end: call, search, answer or hand-over, record and billing.

2. The internship and my role

2.1 The internship

The internship is part of my master's program. It lasts from 7 September to 23 October 2026. I work as the project manager of a real team project and study one area of project management: **finance**.

The internship is finished with three things:

- a **progress video**: project status and the plan for the remaining period;
- an **internship report**, submitted by 23 October 2026;
- a **feedback form** from the company supervisor (mentor).

2.2 The CallMindVoice project

CallMindVoice is a multi-tenant SaaS platform: one system serves many companies, and each company sees only its own data. The target customers are clinics, banks, insurance companies and state service operators in Uzbekistan.

- A company uploads its documents: price lists, FAQs, policies.
- The system answers incoming calls from these documents and names the source.
- When it is not sure, it hands the call to a human operator with a summary and the full transcript.

The team has seven people: two AI engineers (speech; search and AI), two backend engineers (platform; telephony), one frontend engineer, one QA/DevOps engineer, and me. All tasks are given and tracked in our own Jira-like task manager.

2.3 My role: project manager for finance

As a project manager, I did five things:

1. I split the budget into four cost groups: team, servers, AI services and telephony.
2. I estimated the cost of every sprint from the task estimates (story points → hours → money).
3. I compared the planned and the actual cost at the end of every sprint.
4. I made sure that the product itself measures and prices every call, so costs are visible in real time.
5. I compared the traditional way (Excel, monthly reports) with the AI-driven way for my research.

3. The problem

3.1 What is financial management of an IT project?

Financial management means: planning how much the project will cost (the **budget**), checking what it really costs (**cost control**), and making sure the result is worth the money (**ROI**). When nobody controls the money, the project runs **over budget**, and the customer stops trusting it.

3.2 Three finance risks in an AI voice project

Risk	What it means	Example	Result
1. Running costs grow with every call	Phone minutes, speech and AI are paid per minute. More calls means more cost.	A busy clinic with thousands of call minutes a month.	The product can lose money on each call.

Risk	What it means	Example	Result
2. Team hours grow	Hard tasks take longer than planned.	The voice speed work in Sprint 3 took more hours.	The budget is overspent.
3. The value is not visible	The customer pays every month but cannot see what it saves.	"Why do we pay for this?"	The customer leaves.

3.3 Why a spreadsheet alone does not solve it

Many teams keep the budget in Excel. Excel is like a notebook: it **stores** the numbers, but it does not **measure** anything by itself.

Situation	What Excel does	What CallMindVoice adds
A call is made	Nothing. Someone must count the minutes later.	Measures the minutes and multiplies them by the cost per minute at once.
The AI service fails or is expensive	Nothing.	Uses the free local engine first; a paid AI is used only with a key, and on any error the system falls back to the free engine.
The customer asks "what do we save?"	A person must build a report.	The analytics page shows cost, operator hours saved and money saved.
The end of the month	A person prepares the invoice.	Usage is collected every day (usage_daily) and invoices are made from it (invoices).

3.4 Honest note

> The **project budget** numbers in this document (section 7) are **estimates** at market rates, not an accounting report: about 20 hours per person per week, 90,000 UZS/hour for a developer, 100,000 UZS/hour for the PM, 1 USD ≈ 12,700 UZS. The **unit costs** (cost per minute, prices, operator cost) are taken from the project code (src/lib/engine/calls.ts and src/lib/catalog.ts). They are the team's assumptions, not yet checked with a real customer.

4. The solution: how CallMindVoice works and controls cost

4.1 The idea in one sentence

Every call is measured and priced, cheap engines are used first, and a human operator takes over only when the AI is not sure.

4.2 How one call works: four steps

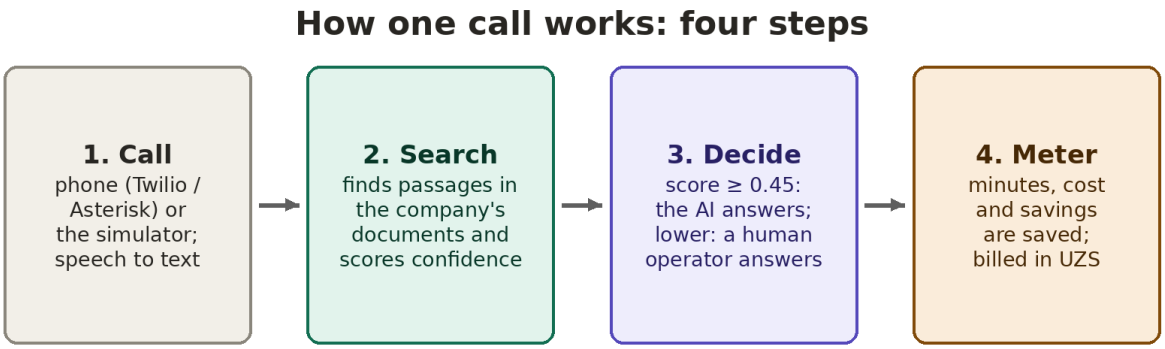


Figure 1. The four steps of one call

Step	Who does it	What happens
1. Call	Phone system or simulator	The call comes through Twilio or an Asterisk phone system. Without them, a built-in simulator makes test calls. Voice detection cuts the audio into phrases; two speech-to-text models (Kotib for Uzbek and Whisper) run in parallel.
2. Search	Program code	The question is matched with the company's documents by keyword search (BM25) and meaning search (vectors) together. The code gives a confidence score.
3. Decide	AI or a person	If the score is 0.45 or more, the language model writes the answer only from the found passages, and it is spoken in Uzbek. If it is lower, or the caller asks for a person, the call goes to an operator.
4. Meter	Program code	The transcript, sources, latency, minutes and cost are saved. Daily usage goes to usage_daily; invoices are made in UZS.

4.3 The finance features

Feature	What it does	Where in the code
Cost per minute	Knows what one minute costs us: telephony $0.018 + speech-to-text $0.004 + AI $0.002 + text-to-speech $0.003 + platform $0.005 = **$0.032** (about 406 UZS).	src/lib/engine/calls.ts (RATE_CARD)
Operator cost model	Compares with a human: a full operator seat costs $650/month, 22 days × 8 hours × 65% busy time = 6,864 call minutes, so **about $0.095 per minute** (about 1,203 UZS).	src/lib/engine/calls.ts (OPERATOR_ASSUMPTIONS)
Saving per AI minute	Operator cost minus our extra cost (telephony is paid in both cases): **about $0.081** per minute that the AI handles alone.	calls.ts (SAVING_PER_DEFLECTED_MINUTE)
Plans and prices	Prepaid balance: 1,000 UZS per AI talk minute. Plans: Trial (500 min free), Pay as you go, Start $199	src/lib/catalog.ts (UZS_RATES, PLANS)

Feature	What it does	Where in the code
	(4,000 min), Pro $890 (25,000 min), Enterprise $2,490 (80,000 min).	
Cheap engines first	A free local AI engine and local embeddings; Claude or Gemini only with a key; automatic fallback on errors.	src/lib/llm/, src/lib/rag/embed.ts
Cached audio	Greetings and holding phrases are stored as ready audio and not created again for every call.	src/lib/speech/tts.ts
Call simulator	The team tests calls without paying for phone minutes.	src/lib/engine/simulator.ts
Analytics	Shows minutes, cost, operator hours saved and money saved.	src/lib/analytics.ts

4.4 What stays with people

- The PM sets the budget, the plan and the priorities.
- The company sets its own confidence threshold and chooses its plan.
- A human operator takes every call that the AI is not sure about.
- Every change in the console is saved in the audit log (who, what, when).

5. What is connected to what

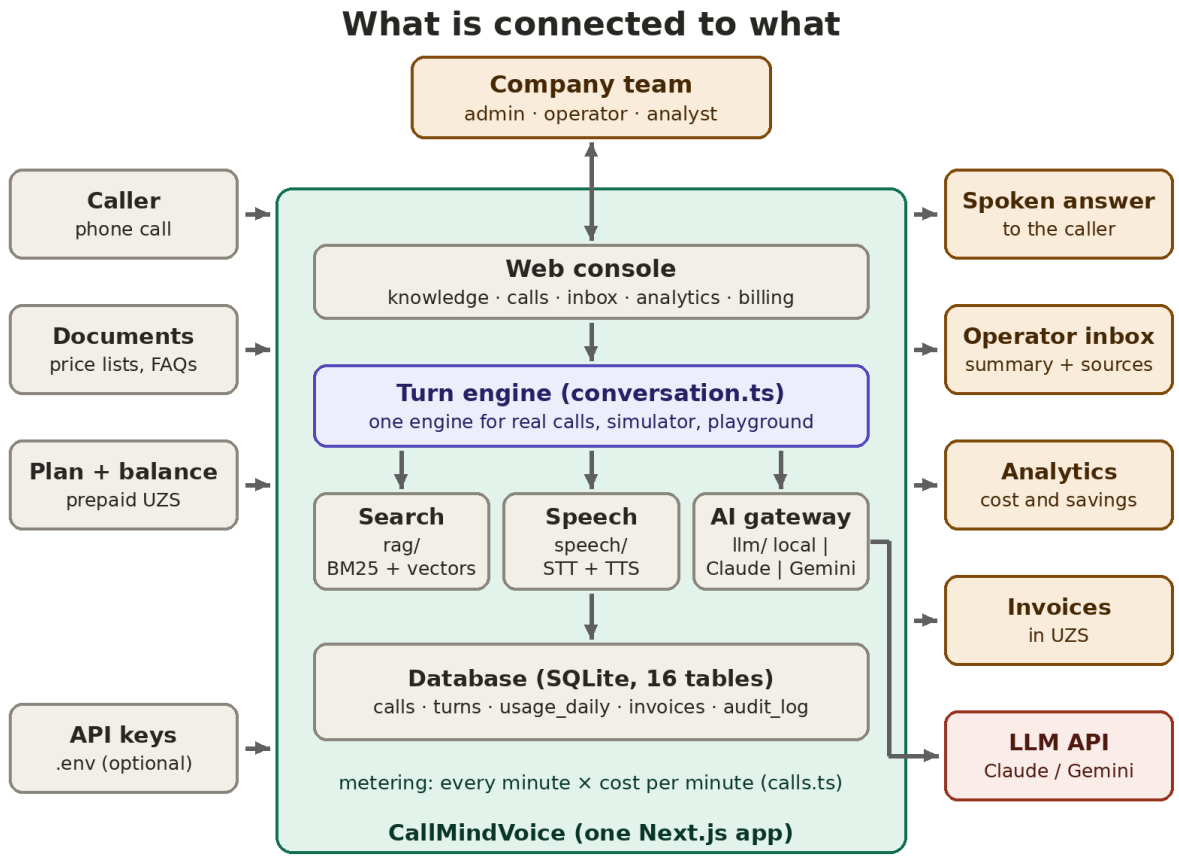


Figure 2. The parts of the system and how they are connected

5.1 The parts of the system

Part	What it does	Connected to	Where in the folder
Web console	The screens for the company team: knowledge, agent, playground, live calls, inbox, analytics, numbers, team, billing, developers, settings.	Company team; turn engine	src/app/app/
Turn engine	Runs one turn of a conversation. Real calls, the simulator and the playground all use it.	Search, speech, AI gateway, database	src/lib/engine/ conversation.ts
Search (RAG)	Splits documents into passages, indexes them, finds the best ones and scores confidence.	Turn engine; documents	src/lib/rag/
Speech	Speech-to-text and text-to-speech, voice detection, cached phrases.	Turn engine; phone	src/lib/speech/, src/app/api/speech/
AI gateway	Chooses the answer engine: local, Claude or Gemini, with fallback.	Turn engine; LLM API; keys	src/lib/llm/
Telephony	Twilio webhook and the Asterisk bridge (a small separate service).	Phone network; turn engine	src/lib/telephony/, telephony-bridge/
Call accounting	Starts and ends calls, counts minutes and cost, writes daily usage.	Turn engine; database	src/lib/engine/calls.ts
Analytics	All the numbers for the dashboards, including cost and savings.	Database; console	src/lib/analytics.ts
Database	SQLite with 16 tables, e.g. calls, turns, usage_daily, invoices, audit_log.	All parts	src/lib/db.ts, data/
Developer API	REST API with API keys and signed webhooks for other systems.	Outside systems	src/app/api/v1/

5.2 Inputs (what goes in)

Input	What it is	Where it comes from
Phone call	The caller's voice.	Twilio, Asterisk, or the simulator.
Documents	PDF, DOCX, TXT, MD, CSV, JSON, HTML files, web pages, FAQs.	The company uploads them in the console.
Plan and balance	The chosen plan and the prepaid balance in UZS.	The billing page.
API keys	Optional keys for Claude, Gemini, OpenAI embeddings, Twilio.	The file .env on the server. Never share it.

5.3 Outputs (what comes out)

- **A spoken answer** to the caller, with the source passages saved.
- **An operator inbox item** when the call is handed over: summary, transcript and sources.

- **Analytics:** answered by AI vs operator, latency, cost, hours and money saved, unanswered questions.
- **Usage and invoices** in UZS, and the audit log.

5.4 The LLM

By default the system uses a **local** answer engine, which costs nothing per call. When a key is set, it uses Claude or Gemini through the internet (an API), with settings for a fast reply. On any error it falls back to the local engine. The answer is always limited to the passages found in the documents.

6. Example: one call from start to end, with its cost

This example uses the built-in clinic knowledge base. The confidence score is the one measured in the project README; the cost uses the numbers from section 4.3 and an average call of 3.3 minutes (the value used in the code).

1. A caller asks in English: "How much is a consultation?"
2. **Search:** the system finds the Uzbek passage "Qabul narxlari" (consultation prices) with a confidence of **0.51**.
3. **Decide:** 0.51 is above the threshold of 0.45, so the AI answers from the passage and names the source.
4. **Meter:** the call lasts about 3.3 minutes. Our cost: 3.3 × $0.032 ≈ **$0.11** (about 1,340 UZS). The company pays 3.3 × 1,000 = **3,300 UZS** from its balance.
5. **The comparison:** a human operator for the same call would cost 3.3 × about 1,203 ≈ **3,970 UZS**. So the company saves money, and we still have a margin.
6. If the question had been harder (for example a score of 0.40), the call would have gone to an operator with a summary, and no AI answer would be guessed.

7. The project budget (estimate)

The budget was split into four cost groups. At the end of every sprint the planned cost was compared with the actual cost. All numbers are **estimates** (see section 3.4).

Cost group	What it includes	Planned (UZS)	Actual (UZS)	Variance (UZS)
Team	PM 110→115 h; 6 developers 660→696 h	70,400,000	74,140,000	−3,740,000 (+5.3%)
Servers	VPS (~$60), nginx/Caddy, HTTPS, backups	800,000	760,000	+40,000
AI services	Claude/Gemini test use	1,900,000	760,000	+1,140,000
Telephony	Twilio test minutes	1,016,000	320,000	+696,000
Total		**74,116,000**	**75,980,000**	**−1,864,000 (−2.5%)**

Indicator	Value and meaning
Budget variance	−2.5%: a little over budget. The team needed more hours for the voice speed work in Sprint 3.
Savings	AI and telephony were about 1.9 million UZS below plan, thanks to the free local engine and the call simulator.

Indicator	Value and meaning
CPI (cost performance index)	About 0.93: about 95% of the planned scope was delivered (EV ≈ 70.4 million UZS, AC ≈ 76.0 million UZS).
Effect of AI coding assistants	Without them, developers would need about 20% more time (~139 hours), so about 12.5 million UZS was saved.
ROI for a customer (example)	A clinic pays 2.5 million UZS a month and saves one operator salary of 6 million UZS: ROI = (6 − 2.5) / 2.5 = 140% a month.

8. Safety and privacy

- **Each company sees only its own data.** Every company table has a tenant_id, and every query filters on it. Roles: owner, admin, operator, analyst.
- **Passwords and sessions are protected:** scrypt password hashing; signed httpOnly session cookies with 30-day expiry.
- **Keys are secret.** API keys are stored only as hashes and shown once; service keys live only in .env.
- **Webhooks are signed:** incoming Twilio webhooks are checked (HMAC-SHA1); outgoing webhooks are signed (HMAC-SHA256).
- **Data stays local** with the local engines: no document text leaves the server.
- **Audit log:** who changed what, and when.

9. How to run it

1. Install Node.js 22.5 or newer.
2. In the project folder, run: npm install
3. Run: npm run dev, and open http://localhost:3000 in the browser.
4. Click **Start free**, sign up, and follow the four-step setup. Click **Load starter knowledge** to get a sample clinic knowledge base.
5. In the console, press **Simulate a call**, then **Backfill 30 days** to fill the analytics with demo data (it is labelled "demo").
6. Open **Analytics** to see the minutes, cost and savings, and **Billing** to see the plan, balance and invoices.

For a server: npm run build and npm start, behind nginx or Caddy with HTTPS. No API keys are needed; every engine has a free local version.

10. Testing and results

10.1 What was checked

What was checked	Result
Type check of the whole code (npm run typecheck)	Used as the main automatic check
Search on the clinic knowledge base, 4 questions in 3 languages	All 4 found the right passage; confidence 0.47–0.84
Full call path with the simulator	Works: call → search → answer or hand-over → record → analytics

What was checked	Result
Voice reply speed target	Under about 800 ms to the first audio (the design target of Sprint 3)
MVP status	Ready on 16 October 2026

> **Honest note.** The project has no automatic test suite yet; checks were the type check, the measured search examples and manual call scenarios with the simulator. The money numbers in section 7 are estimates. The next step is a pilot with a real clinic, to check the real cost per call and the real savings.

11. Traditional vs AI-driven financial management

This section connects the project with my research topic.

Stage	Traditional approach	AI-driven approach	Example in CallMindVoice
Cost estimation	Expert judgement; hours × rate in Excel	Cost is predicted from the data of past sprints	Story points were turned into hours and money; AI assistants sped up estimation
Budgeting	The budget is made once at the start	The budget is updated every sprint from real data	Planned and actual cost compared at the end of every sprint
Cost control	Monthly reports; problems are found late	Real-time tracking and early warnings	Every call is metered; usage is written every day
Cost optimisation	A person decides which service is cheaper	The system chooses the cheaper engine by itself	Local engine first, fallback, cached audio, simulator
Labour cost	Work is split by the PM's experience	AI assistants write routine code	About 12.5 million UZS of hours saved (estimate)
Value (ROI)	Measured at the end of the project	Cost and savings are calculated all the time	Analytics shows money saved per AI-handled minute

Main finding: AI-driven systems make financial management faster and more accurate, because the costs are measured all the time and cheaper options are chosen automatically. But the budget, the priorities and the final decisions still belong to people.

12. Limits and next steps

Limit today	Next step
The budget numbers are estimates.	Replace them with real hours and real bills.
Unit costs are the team's assumptions.	Check the real cost per call in a pilot with a clinic.
Plan changes are recorded but not charged.	Connect Payme, Click or Stripe.
Only 3 of 11 industry starter packs have real content.	Replace the sample content before pilots.
Scanned PDFs cannot be read (no OCR).	Add OCR.
One server only (SQLite, in-process events).	Move to Postgres + pgvector and Redis when needed.
No automatic test suite.	Add tests for search, billing and cost calculation.

13. Where to find everything

Folder / file	What is inside
github.com/Kaizen-is/CallMindVoice	The project repository.
README.md	What is built, how to run, engines, security.
PROJECT_BRIEF.md	Project brief for developers and AI assistants: what is real, what is simulated, known defects.
src/app/	The website, the console and the APIs.
src/lib/engine/calls.ts	Call accounting: cost per minute, operator cost model, savings.
src/lib/catalog.ts	Plans, prices in USD and UZS.
src/lib/analytics.ts	All dashboard numbers, including cost and savings.
src/lib/rag/, src/lib/llm/, src/lib/speech/	Search, AI answer engines, speech.
telephony-bridge/	The separate Asterisk bridge service.
data/	The database and uploads on the server (back it up).

14. Glossary (simple words)

Word	Meaning
Budget	The planned amount of money for the project.
Cost control	Checking the real costs against the budget.
Variance	The difference between the planned and the actual cost.
CPI	Cost performance index: the value of the work done divided by the real cost. Below 1 means over budget.
EV / AC	Earned value (value of the work done) / actual cost.
ROI	Return on investment: (money saved − money spent) / money spent.
Cost per minute	What one minute of a call costs us.
Deflection	A call that the AI handles alone, without an operator.
Multi-tenant	One system that serves many companies, each with its own data.
RAG	Retrieval-augmented generation: the AI answers only from found passages.
BM25 / vectors	Keyword search / meaning search.
Confidence score	A number that shows how sure the system is about the answer.
STT / TTS	Speech-to-text / text-to-speech.
LLM	Large language model: a program that understands and writes text.
API	A way for one program to talk to another over the internet.
Fallback	Using a backup option when the main one fails.
Story points	A number for the size of a task.
MVP	Minimum viable product: the first working version.
Audit log	A list that saves who changed what, with the time.